# GROUP: FD2

## PRODUCT REQUIREMENT DOCUMENT (PRD)

## UMHackathon 2026

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**Project Name:** Upstart

**Version:** 1.0

## 1. Project Overview

- **Problem Statement:**
  The urban poor in Malaysia (specifically the B40 group) live in a high-stress "financial fog." While digital banking and sophisticated budgeting apps exist, they are designed for salaried middle-class users with digital footprints.
- **Target Domain:**
  Financial Inclusion & Personal Finance Management (PFM) for the Informal Urban Economy Group.
- **Proposed Solution Summary:**
  **Upstart** is a "Financial Survival Co-pilot" that strips away the complexity of traditional accounting in favor of AI-driven reasoning.
  - **Human-Centric Logging:** A mobile-first interface optimized for ultra-fast cash logging (tap-to-log) and voice-to-text entries, removing the friction of manual data entry.
  - **The "Logic Brain" (GLM by Z.AI):** Instead of just charts, the AI provides plain-language directives. It acts as a supportive peer that analyzes income volatility and debt structures to suggest realistic daily spending limits.
  - **Behavioral Debt Strategy:** The system automatically categorizes debt by urgency and interest, offering the user a choice between the Snowball Method (for psychological motivation) or the Avalanche Method (for mathematical efficiency).

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- **The Business Bridge:** Unlike standalone budgeters, this project connects personal survival to income growth. If the AI detects a persistent deficit, it triggers the Business Layer to help the user professionalize their side-hustle (e.g., Auntie Siti's nasi lemak) to bridge the gap.
- **Gamified Health Score:** Converts complex financial ratios into a single, color-coded Financial Health Score (0-100) to provide immediate, non-judgmental feedback on progress.

## 2. Background & Business Objective

- **Background of the Problem:**
  The Informal Urban Economy Group comprising B40 households, gig workers, and micro-vendors in Malaysia's cities operates within a persistent "financial fog" caused by a heavy reliance on cash transactions that leave no digital footprint. Unlike salaried employees who benefit from automated banking insights, this group's financial life is invisible to traditional analytics, making it nearly impossible to track spending or predict savings. This lack of visibility is compounded by high income volatility and a reliance on informal credit markets, such as "Buku 555" store credit or high-interest informal lenders, which frequently trap families in "debt spirals" where they struggle to cover interest while the principal remains untouched. Without a clear mechanism to visualize their cash-based reality, these individuals are often one minor emergency away from total financial collapse.

- **Importance of Solving This Issue:**
  Addressing these challenges is critical for building financial resilience and breaking the cycle of generational poverty within urban centers. For a family living hand-to-mouth, a single "economic shock" like a vehicle breakdown or a sudden medical bill can force them into predatory borrowing, creating a downward shift in quality of life that takes years to reverse. By providing a tool that simplifies complex financial logic into actionable daily steps, we can reduce the immense "cognitive load" and anxiety

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associated with scarcity. Solving this issue transforms financial management from a source of stress into a tool for empowerment, allowing users to build a small but vital savings buffer that acts as a shield against the volatility of the informal economy.

- **Strategic Fit / Impact:**
  **Upstart** serves as a strategic bridge for financial inclusion, aligning with national goals to digitalize the informal sector and prepare them for formal financial services. By utilizing a Generative Language Model (GLM), the project provides "Private Banking" level advice to the underserved, offering a scalable solution that speaks their language and understands their specific socio-economic constraints. The strategic impact lies in the "Business-Finance Bridge", as users master their survival budget in the Finance Layer, the system identifies opportunities to scale their side-hustles in the Business Layer. Ultimately, this creates a data-driven path for upward mobility, turning unbanked micro-entrepreneurs into resilient, documented, and financially literate contributors to the broader economy.

## 3. Product Purpose

### 3.1. Main Goal of the System:

To provide a low-friction, AI-powered "survival toolkit" that helps the Informal Urban Economy Group clear the "financial fog" of cash-based living through actionable, plain-language guidance. By simplifying debt management and expense tracking, the system empowers users to build financial resilience and bridge the gap from daily survival to sustainable income growth.

### 3.2. Intended Users (Target Audience):

- Micro-Entrepreneurs
- Variable Income Workers
- Cash Users
- Simple Phone Users
- Debt-Burdened Individuals

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## 4. System Functionalities

### 4.1. Description:

Upstart acts as a proactive financial co-pilot that uses high-speed cash logging and AI reasoning to turn "invisible" cash transactions into actionable survival strategies. The system shifts the burden of analysis from the user to the AI, delivering plain-language advice on debt reduction, emergency savings, and opportunities to increase income.

### 4.2. Key Functionalities:

- **Vibe-Based UX Design:** The interface is designed with fast, thumb-friendly interactions and clear visual hierarchy to make financial tracking feel simple, supportive, and low-stress for users managing daily cash-based finances.
- **Automated Financial Intelligence:** The system uses Z.AI GLM to convert unstructured financial inputs (text) into structured data, automatically categorizing income, expenses, and debt while identifying key financial patterns and insights.
- **Dynamic Financial Insights Engine:** The platform continuously analyzes user financial data to generate actionable insights such as spending trends, risk detection, and savings opportunities, helping users understand their financial situation without manual analysis.
- **Component-Based Data Structuring (Planned Feature):** Financial data is organized into modular cards such as income summaries, expense breakdowns, and debt overviews to support future extensibility. In the current MVP, these components are used for in-app visualization only. Export and sharing capabilities for individual financial modules (e.g. debt reports for micro-loan applications or family sharing) are not implemented in the current version but are planned for future development.

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### 4.3. AI Model & Prompt Design

#### 4.3.1. Model Selection:

The system utilizes Z.AI GLM as its core intelligence engine because of its superior ability to process "messy" unstructured data such as informal messages and irregular income logs, and translate them into structured JSON records for the app's database. Its advanced reasoning capabilities make it the ideal choice for this project, as it can connect disparate financial data points (e.g., rising food costs and stagnant income) to deliver empathetic, plain-language advice that feels like a human conversation rather than a cold calculation. Furthermore, the model's efficiency in low-bandwidth environments ensures that users in the Informal Urban Economy Group receive near-instant financial guidance without requiring high-end hardware or expensive data plans.

#### 4.3.2. Prompting Strategy:

Our team utilizes a Multi-Step Agentic Prompting approach to handle the complexity of the informal urban economy. This strategy was chosen because financial survival for the B40 group is not a "one-shot" problem. It requires the AI to first retrieve full financial context, reason through conflicting priorities (like choosing between buying school supplies or paying a high-interest debt), and then execute a personalized plan.

#### 4.3.3. Context & Input Handling:

The system handles unstructured financial data by using the Z.AI GLM to extract structured financial entities from informal text, ensuring that real-world, messy inputs are accurately categorized into income, expenses, and debt records. This allows users to log financial information in natural language while still maintaining structured and usable financial data for analysis.

The system supports a large input capacity of approximately ~10,000–20,000 tokens per session depending on model configuration to accommodate extended financial histories and continuous user activity. If this limit is exceeded, the system applies a chunking and summarization process that progressively compresses older or redundant information into high-level financial

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summaries. This ensures that key financial insights are preserved while maintaining system performance and preventing data rejection.

#### 4.3.4. Fallback & Failure Behavior:

When the GLM model produces off-topic, unusable, or potentially hallucinated responses, the system applies a validation layer before displaying any output to the user. Each response is checked against real financial data from system tools (income, expenses, debt records) to ensure consistency, correctness, and proper grounding. If the output is found to be inconsistent or unreliable, the system automatically triggers a single retry attempt using a more constrained and structured prompt to improve accuracy.

If the retry still fails validation, the system will not display AI-generated advice and will instead switch to a graceful fallback state. In this state, the user is shown only deterministic financial information such as summaries of income, expenses, and debt, along with a clear message indicating that reliable advice cannot be generated at the moment. In repeated failure cases, the AI advisor may be temporarily restricted to safe mode, where only rule-based calculations are provided without generative recommendations, ensuring that no hallucinated financial advice is presented to the user.

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**5. User Stories & Use Cases**

- **User Stories:** *"As an informal vendor, I want to log my daily cash sales using my own words so that the system can automatically categorize my income and update my financial overview without requiring me to use complex spreadsheets or manual bookkeeping tools."*
- **Use Cases (Main Interactions):**

  **Prompt to UI Flow:** Users enter financial updates or goals in natural language. The Z.AI GLM processes these inputs to extract structured financial entities such as income, expenses, debt, or savings goals. The system then converts this structured data into modular financial cards that are displayed on the dashboard, providing a clear and updated visual summary of the user's financial status.

  **Flow Generation:** When a user interacts with a financial module or selects a specific action (e.g. viewing debt breakdown or requesting savings advice), the system triggers a contextual flow generation process. GLM interprets the user's intent and current financial state to determine the most relevant next view. It then dynamically updates or transitions between modules such as Dashboard, Debt Planner, and AI Advisor while maintaining consistent data state and UI structure. This ensures seamless navigation without losing financial context or requiring manual recalculation of information.

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**6. Features Included (Scope Definition)**

- Conversational financial input system where users can describe income, expenses, debt, or savings goals in natural language, which is then converted into structured financial records.
- Real-time AI Financial Advisor powered by GLM, which analyzes user financial data and generates actionable, context-aware recommendations.
- Consistent design system across all modules, ensuring unified layout, tokens, and synchronized financial state throughout the application.
- Dynamic flow generation between modules (Dashboard, Debt, Debt Planner, Savings Goals, AI Advisor), allowing seamless navigation without loss of context or data inconsistency.
- Instant data reflection across the system, where any updates to financial inputs are immediately updated across all relevant views and AI outputs.

**7. Features Not Included (Scope Control)**

- Full production-grade backend infrastructure such as scalable microservices architecture and distributed systems.
- Complex financial integrations such as banking APIs, real-time transaction syncing with banks, or credit bureau connections.
- Multi-user collaborative financial editing or simultaneous shared accounts in real time.
- Full offline support or cross-device real-time synchronization across multiple devices.
- Automated investment trading, stock portfolio management, or wealth management functionalities.
- External payment processing or direct fund transfer capabilities within the application.

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**8. Assumptions & Constraints**

- **LLM Cost Constraint:** The system assumes an average user session consumes approximately 2,000–4,000 tokens, depending on the number of financial queries and AI advisory interactions. To manage inference cost at scale, the system implements response caching for repeated financial summaries and uses structured tool-based retrieval (income, expenses, debt summaries) to reduce redundant LLM calls.
- **Technical Constraints:** The system is built as a web-based MVP with support limited to frontend and lightweight backend logic. It does not include full production-grade infrastructure such as mobile-native applications (Swift for iOS or Kotlin for Android) or advanced framework exports (e.g. full React app generation or native deployment code).
- **Performance Constraints:** Due to reliance on LLM-based financial analysis and UI generation, response generation is subject to token and latency limits. Large financial histories are processed using chunking and summarization to prevent excessive token usage and maintain acceptable response times.
- **User Input Constraints:** The system depends on user-provided financial data for analysis and cannot operate autonomously without input. Outputs are generated based on available data, and in cases of missing or incomplete information, the system prompts users for clarification rather than generating assumptions.

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**9. Risks & Questions Throughout Development**

- **AI Hallucination in Financial Advice:** There is a risk that the GLM model may generate inaccurate or ungrounded financial recommendations. This is mitigated through a validation layer that cross-checks AI outputs against structured financial data (income, expenses, debt) before displaying results to the user.
- **Data Inconsistency Across Modules:** Since multiple modules (Dashboard, AI Advisor, Debt, Savings) rely on shared financial data, there is a risk of mismatch or outdated state representation. This is addressed through centralized state management and real-time synchronization of financial records across all components.
- **Performance Under Large Financial Histories:** As user financial data grows, there may be latency in processing or generating AI insights. This is managed through chunking, summarization, and caching of previously computed financial summaries to reduce repeated computation.
- **User Input Ambiguity:** Users may input vague or incomplete financial information, leading to unclear categorization by the AI. The system handles this by prompting for clarification or assigning low-confidence labels that require user confirmation before final storage.
- **Cost and Token Overuse:** High-frequency AI queries may increase operational cost due to token usage. To mitigate this, the system limits redundant AI calls by reusing cached results and prioritizing deterministic calculations for simple financial summaries.

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